

JV Food

A/A/B Test Analysis

User Behavior & Funnel Insights

7,534

Users

240K

Events

5

Funnel
Stages

Tools used:

Python

Pandas

SciPy

StatsModels

Matplotlib

Seaborn

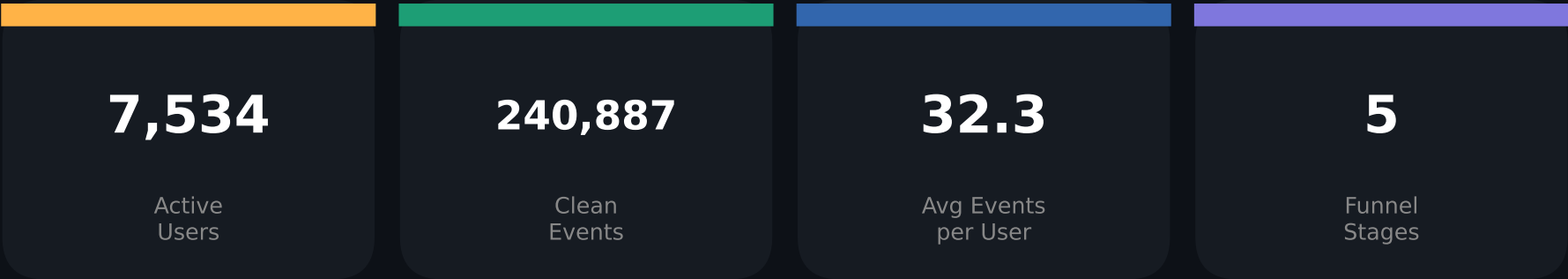
TripleTen Data Analyst Bootcamp — Sprint 9

The Dataset

What we worked with

Period: July 25 - Aug 7, 2019 (13 days)

Records before Aug 1st removed (0.99%) -> cleaner, more complete data



Funnel stages:



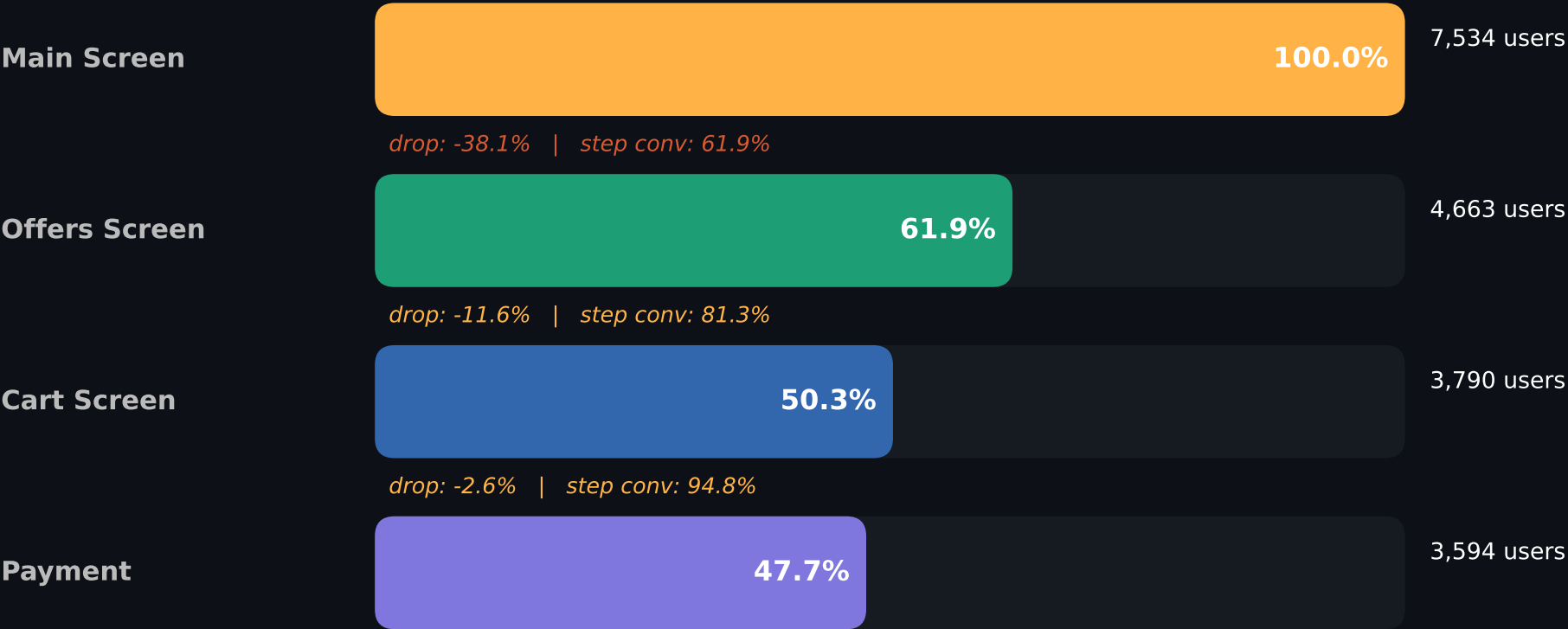
Power Users detected:



-> Large gap between P75 and P99 signals a highly engaged power-user segment.

The Funnel

Where users go — and where they leave



[!] BIGGEST DROP: Main Screen -> Offers = 38.1% of users lost

The friction is NOT at checkout. Users drop before they even see what's for sale.

[OK] Once past Offers: 81.3% -> Cart and 94.8% -> Payment

The A/A/B Test

Experiment design & validation

Group 246

Control A

2,537 users

Group 247

Control B

2,534 users

Group 248

Treatment

2,463 users

What was tested: font change in Group 248 vs. two identical control groups

The A/A design validates that both controls are statistically equivalent before comparing to treatment

A/A Validation Result:

[OK] Z-test: Groups 246 & 247 are statistically equivalent

Experiment baseline is solid. Both controls behave identically.

Statistical Methodology:

- **Z-test for proportions**

— compare conversion rates at each funnel stage per group

- **12 total comparisons**

— 4 stages x 3 group pairings (246/247, 246/248, 247/248, combined/248)

- **Holm-Bonferroni correction**

— controls Type I error (false positives) across multiple tests

The Results

All 12 tests — after Holm-Bonferroni correction

Comparison	Stage	Orig. p-value	Corrected p	H0 Rejected?
246 vs 247	Offers	0.262	1.0	No
246 vs 247	Cart	0.240	1.0	No
246 vs 247	Payment	0.120	1.0	No
246 vs 248	Offers	0.268	1.0	No
246 vs 248	Cart	0.101	1.0	No
246 vs 248	Payment	0.258	1.0	No
247 vs 248	Offers	0.987	1.0	No
247 vs 248	Cart	0.641	1.0	No
247 vs 248	Payment	0.670	1.0	No
Control vs 248	Offers	0.531	1.0	No
Control vs 248	Cart	0.224	1.0	No
Control vs 248	Payment	0.586	1.0	No

[!!!] All 12 corrected p-values = 1.0 -> H0 not rejected in ANY case

The font change had zero measurable impact at any stage of the funnel.

The Takeaway

What this means for the product

[OK] The font change had no effect.

Statistically confirmed across 12 tests with Holm-Bonferroni correction.

The Real Opportunity:

38.1% of users NEVER see the offers.

Fix the Main Screen -> Offers transition and you unlock the biggest revenue lever.

TEST 1

Redesign the Main Screen CTA to drive Offer exploration

TEST 2

A/B test personalized offer banners based on user history

TEST 3

Push notifications triggered after Main Screen with no Offer click

Full project on GitHub | Python · Pandas · SciPy · StatsModels · Matplotlib · Seaborn

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